CS 456 - Mathematical Foundations of Machine Learning

Professor: Erisa Terolli
Semester: Spring 2026

Due Date: February 9 2026, 11:59 PM
Assignment: Number 1

Submission Write your solutions in LaTeX and submit your homework as a PDF file on Canvas. To receive full credit, make sure to show all steps and provide clear explanations for your work.

Problem 1 (20 points) Solve the following system of linear equations by elimination and back substitution:

$$x + 2y + 3z = 0$$
$$-x + y - 2z = -3$$
$$2x + y + z = 3$$

$$\begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & -2 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \xrightarrow{\substack{R2 = R2 + R1 \\ R3 = R3 - 2R1}} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \\ 0 & -3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \xrightarrow{R3 = R3 + R2}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix}$$

$z = 0$
$y = -1$
$x = 2$

$$\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$$

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Problem 3 $\begin{bmatrix} -3 \\ -6 \\ 3 \end{bmatrix}$, and

Problem 2 (15 points) Let A be any nxn matrix.

(a) If A is invertible, show that A^2 is invertible.

(b) Suppose A^2 is invertible. Can A be singular? If yes, give such an example. If not, explain why A is non-singular.

a) Matrix A is invertible → A^{-1} exists

A^2 is invertible if there exist a matrix, that when multiplied by A^2 will give I.

$A^2(A^{-1}A^{-1}) = A(A\,A^{-1})A^{-1} = A(I)A^{-1} = A\,A^{-1} = I$

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Problem 3 (25 points) Express $x = \begin{bmatrix} 1 \\ -2 \\ 7 \end{bmatrix}$ as a linear combination of $a = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}$, $b = \begin{bmatrix} -3 \\ -6 \\ 3 \end{bmatrix}$, and $c = \begin{bmatrix} 4 \\ 10 \\ 10 \end{bmatrix}$.

$$\begin{bmatrix} 2 & -3 & 4 \\ 4 & -6 & 10 \\ 4 & 3 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 7 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 2 & -3 & 4 & 1 \\ 4 & -6 & 10 & -2 \\ 4 & 3 & 10 & 7 \end{array}\right] \xrightarrow[R3=R3-2R1]{R2=R2-2R1} \left[\begin{array}{ccc|c} 2 & -3 & 4 & 1 \\ 0 & 0 & 2 & -4 \\ 0 & 9 & 2 & 5 \end{array}\right] \xleftrightarrow{R3 \leftrightarrow R2} \left[\begin{array}{ccc|c} 2 & -3 & 4 & 1 \\ 0 & 9 & 2 & 5 \\ 0 & 0 & 2 & -4 \end{array}\right]$$

$z = -2$

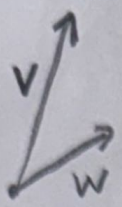
$9y - 4 = 5 \rightarrow y = 1$

$2x - 3y + 4z = 1 \rightarrow 2x - 3 - 8 = 1 \rightarrow x = 6$

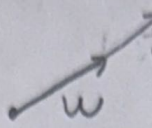
$$6\begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} + 1\begin{bmatrix} -3 \\ -6 \\ 3 \end{bmatrix} - 2\begin{bmatrix} 4 \\ 10 \\ 10 \end{bmatrix} = \begin{bmatrix} 12 - 3 - 8 \\ 24 - 6 - 20 \\ 24 + 3 - 20 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 7 \end{bmatrix} \checkmark$$

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Problem 4 (10 points) If $||v|| = 5$ and $||w|| = 3$, what are the smallest and largest values of $||v - w||$? What are the smallest and largest values of $v \cdot w$?

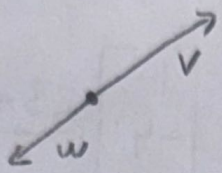


Smallest value is



$\rightarrow ||v - w|| = 2$

Largest value



$\rightarrow ||v - w|| = 8$

$v \cdot w = ||v|| \, ||w|| \cos\theta \qquad \cos\theta \cdot [-1, 1]$

Smallest -15, Largest 15

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Problem 5 (15 points) Find the determinant of matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 3 & 2 \\ 1 & -1 & -2 \end{bmatrix}$.

$$\det\left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 3 & 2 \\ 1 & -1 & -2 \end{bmatrix}\right) = 1\begin{vmatrix} 3 & 2 \\ -1 & -2 \end{vmatrix} + 1\begin{vmatrix} 0 & 3 \\ 1 & -1 \end{vmatrix}$$

$$= -4 - 3 = -7$$

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Problem 6 (15 points) Use Gauss-Jordan elimination to find A^{-1} if $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 1 & 0 & 1 \end{bmatrix}$.

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \rightarrow \text{Do eliminations on both sides}$$

$$A^{-1} = \begin{bmatrix} -1 & 0 & 2 \\ -1 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

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